Machine Learning-Based GHI Prediction and Web Deployment Using Leave-One-Station-Out Cross-Validation in Saudi Arabia

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Abstract

Solar radiation (SR) serves as a primary energy source for solar-based renewable energy systems. Accurate prediction of SR is essential for the effective planning, operation, and utilization of solar energy systems. This study presents a machine learning (ML) framework for predicting Global Horizontal Irradiance (GHI) using meteorological and SR data collected from 43 monitoring stations across Saudi Arabia. A unified preprocessing pipeline is applied, including missing-value treatment, normalization, and feature scaling to improve model consistency and predictive performance and compares performance with eight ML algorithms. The ML models are evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), coefficient of determination ($\boldsymbol{R^2}$), and training time. Station-based Leave-One-Group-Out and 10 fold cross-validation are used fo robustness and geographical generalization of models.

The experimental results demonstrate that Histogram-based Gradient Boosting (HGB) achieved the best overall predictive performance, obtaining the lowest MSE and RMSE, and the highest $\boldsymbol{R^2}$. Linear Regression (LR) achieved the lowest MAE while requiring only 0.01 s of training time, making it the most computationally efficient and interpretable model. The best-performing model was further integrated into a lightweight web-based application for interactive station visualization and real-time GHI prediction. he proposed framework demonstrates the effectiveness of ML for reliable GHI prediction and provides a computationally efficient approach for solar energy assessment across geographically distributed monitoring stations.

Keywords: Solar Radiation Prediction, Global Horizontal Irradiance (GHI), Machine Learning, Leave-One-Station-Out Cross-Validation, Renewable Energy, Saudi Arabia, Meteorological Data, Comparative Analysis

1 Introduction

Solar energy has become an important renewable energy source as countries move towards cleaner and more sustainable forms of energy. The ability to predict solar radiation (SR), particularly Global Horizontal Irradiance (GHI), is important for the planning and operation of solar energy systems because changes in available solar radiation directly affect expected energy production. Reliable solar radiation prediction can therefore support the planning and operation of photovoltaic and other solar energy systems [1, 2].

Saudi Arabia has favorable conditions for solar energy due to its high and relatively stable solar irradiance. At the same time, the country's large geographical area results in variations in local environmental conditions that can influence solar radiation. Parameters such as temperature, humidity, wind speed, Direct Normal Irradiance (DNI), Diffuse Horizontal Irradiance (DHI), and atmospheric pressure provide useful information for estimating GHI.

Traditional approaches to SR prediction generally rely on empirical relationships, statistical methods, or physical modeling. Machine learning (ML) provides an alternative approach by learning relationships between observed environmental variables and solar radiation from historical data. Previous studies have investigated a ML techniques for SR prediction, including artificial neural networks and other data-driven approaches [1–3]. More recent studies have also explored ensemble and hybrid approaches for improving prediction performance [4, 5].

In this work, a ML framework is developed for predicting GHI using data collected from 43 geographical monitoring stations across Saudi Arabia. The dataset contains solar radiation and meteorological measurements, including GHI, DNI, DHI, temperature, humidity, wind speed, and pressure. Eight ML algorithms are evaluated: Linear Regression (LR), Support Vector Regression (SVR), Decision Tree (DT), Random Forest (RF), K-Nearest Neighbors (KNN), XGBoost, Histogram-based Gradient Boosting (HGB), and Artificial Neural Network (ANN).

The ML model performance and generalization using 10 fold and Leave-One-Group-Out cross-validation, with each monitoring station treated as a group. In this evaluation, one station is excluded during training and used for testing, allowing the performance of the models across different geographical locations to be examined.

In addition to the ML models, a web-based application was developed to make the prediction framework accessible through an interactive interface. The application uses Flask for the backend and integrates Leaflet.js and Plotly.js for map-based and graphical visualization. Users can interact with station information and obtain GHI predictions through the application.

The main contributions of this work are summarized as follows:

- Development of a robust and geographical generalization of machine learning models for predicting Global Horizontal Irradiance (GHI) using solar radiation and meteorological observations collected from 43 monitoring stations across Saudi Arabia.
- Comparative evaluation of eight machine learning algorithms using common regression performance metrics.
- Evaluation of model performance using 10 fold and station-based Leave-One-Group-Out cross-validation to examine model behavior across different geographical locations.
- Development of a lightweight web-based application using Flask, Leaflet.js, and Plotly.js for interactive station visualization and real-time GHI prediction.

2 Related Work

SR prediction has been studied using statistical, physical, and machine learning approaches because of its importance in solar energy planning and system operation. Among data-driven methods, machine learning has received considerable attention because it can learn relationships between solar radiation and meteorological variables without requiring an explicit physical model of the atmosphere.

Yadav and Chandel [3] reviewed the use of artificial neural networks (ANNs) for SR prediction and discussed their ability to model nonlinear relationships between meteorological inputs and solar irradiance. Qazi *et al.* [1] provided a systematic review of ANN-based methods for SR prediction and solar system design, covering applications across different geographical and climatic conditions. These studies established the usefulness of neural network models for learning complex relationships in SR data.

The use of different ML strategies has also been examined from a broader perspective. Zhou *et al.* [2] reviewed global SR prediction using ML models and discussed the influence of input variables, data preprocessing, feature selection, and model evaluation on prediction performance. Guermoui *et al.* [4] focused on hybrid approaches for solar radiation forecasting and showed how combining different modelling techniques can be useful for handling the variability of solar radiation data.

Some recent work has explored ensemble learning and the use of additional data sources. Al-Shourbaji *et al.* [5] investigated the use of Random Forest with the Cheetah Optimizer for SR prediction, showing the potential of optimized ensemble methods for this task. Attya *et al.* [6] combined satellite imagery with tabular data for solar radiation prediction, highlighting the possibility of improving prediction systems by incorporating different types of information. Yadav *et al.* [7] compared deep learning models for daily global solar radiation prediction using stochastic input variables, further demonstrating the application of neural models to solar radiation forecasting.

Although these studies demonstrate that ML can be effective for SR prediction, the performance of a model can vary with the dataset, geographical region, input variables, and evaluation procedure. A direct comparison of several commonly used supervised learning algorithms under the same preprocessing and evaluation conditions can therefore provide useful insight into their relative performance. In addition, when observations are collected from multiple monitoring stations, evaluating a model on data from a station that was not used during training provides an additional way to examine its behavior across different geographical locations.

3 Proposed Methodology

This section describes the methodology followed for predicting GHI from SR and meteorological observations collected across Saudi Arabia. The complete process includes data collection, preprocessing, selection of the input variables, model training, performance evaluation, cross-validation, and deployment of the trained model through a web-based application. The overall workflow of the study is shown in Fig. 1.

The collected data first cleaned to handle incomplete records and non-informative attributes. The selected numerical variables were then normalized and used as inputs for the ML models. The ML models were trained and evaluated using the same prepared dataset. Model performance was evaluated using MAE, MSE, RMSE, R^2, and training time. Finally, the trained models were integrated into a web-based application for geographical and graphical visualization. This provides an interactive way to explore station information and obtain GHI predictions from the developed system.

3.1 Dataset Description

The dataset used in this study consists of SR and environmental observations collected from monitoring stations across Saudi Arabia. The Atomic and Renewable Energy Department at King Abdullah City in Saudi Arabia collects and maintains a publicly available SR prediction dataset. The dataset is available on the OpenData platform, a Saudi Arabian government repository atb different geographical locations. Each record contains 26 features, including SR measurements, meteorological variables, and additional attributes related to the recorded observations.

The main target variable in this study is GHI, which represents the total SR received on a horizontal surface. Table 3.1 summarizes the input features employed for GHI prediction along with their corresponding minimum and maximum values. These variables were selected because they provide the information required by the prediction models to estimate GHI.

The observations were recorded on an hourly basis. After removing incomplete and unusable records during the data preparation stage, the final dataset contained 1,265 usable observations. The number of observations was not uniform across all monitoring stations, with some stations contributing only a small number of records while others provided substantially more observations. This variation resulted in differences in the amount of data available from each geographical location.

The dataset also contains additional attributes such as station identifiers, location information, timestamps, measurement uncertainty, and quality-related fields. These

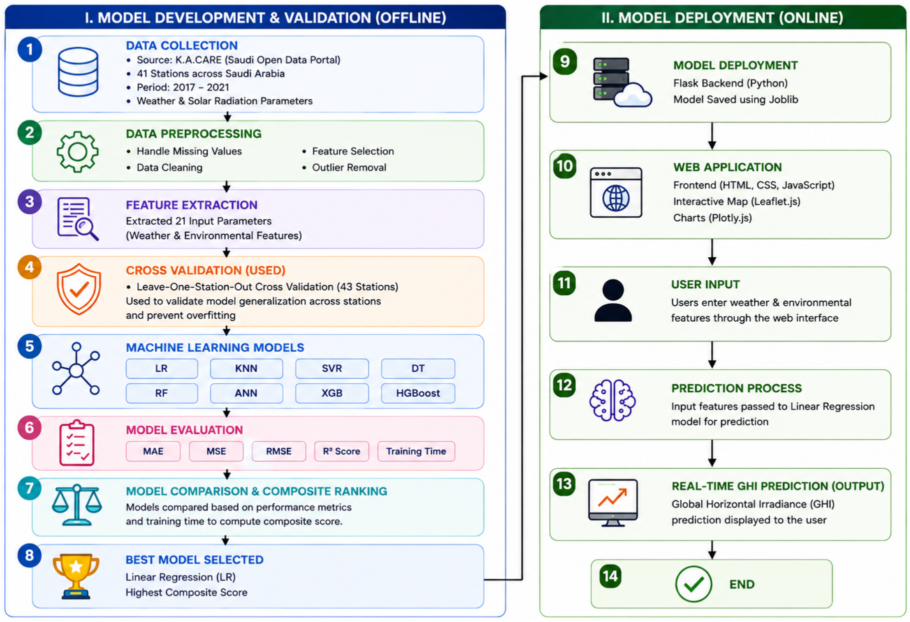


Fig. 1 Overall workflow of the proposed ML model development and deployment in web application for solar radiation prediction.

Table 1 Statistical summary of Saudi Arabia SR dataset.

Parameter	Abbr.	Unit	Min	Max
Ambient Air Temperature	AT	Celsius	−2.546	1.810
Air Temperature Error Range	ATU	Celsius	−0.040	25.100
Atmospheric Pressure	BP	hPa	−2.657	1.431
Atmospheric Pressure Error Range	BPU	hPa	−2.459	4.662
Diffuse Horizontal Solar Radiation	DHI	Wh/m^2	−1.864	2.692
Diffuse Horizontal Radiation Error	DHIU	Wh/m^2	−1.629	8.167
Diffuse Horizontal Radiation Variability	DHISD	Wh/m^2	−2.396	4.263
Direct Normal Solar Radiation	DNI	Wh/m^2	−2.900	3.089
Direct Normal Radiation Error	DNIU	Wh/m^2	−1.767	6.931
Direct Normal Radiation Variability	DNISD	Wh/m^2	−2.659	3.003
Global Horizontal Radiation Error	GHIU	Wh/m^2	−0.729	19.933
Global Horizontal Radiation Variability	GHISD	Wh/m^2	−1.790	4.114
Maximum Wind Velocity at 3 m	PWS	m/s	−3.978	4.746
Maximum Wind Velocity Error at 3 m	PWSU	m/s	−5.274	5.507
Ambient Relative Moisture	RH	%	−1.494	2.269
Relative Moisture Error Range	RHU	%	−8.375	20.765
Wind Azimuth at 3 m	WD	Degree	−1.542	1.447
Wind Direction Error at 3 m	WDU	Degree	−13.497	2.896
Wind Velocity at 3 m	WS	m/s	−3.271	5.054
Wind Velocity Error at 3 m	WSU	m/s	−1.549	0.678
Wind Velocity Variability at 3 m	WSUSD	m/s	−3.761	4.954
Global Horizontal Solar Irradiance	GHI	Wh/m^2	0.000	1.000

attributes were retained during the data collection stage but were not used as predictive variables in the final ML models. The selected input variables present in Table 3.1, for prediction were temperature, humidity, pressure, wind speed, DNI, and DHI, while GHI is used as the target variable.

The geographical distribution of the observations provides data from different environmental conditions across Saudi Arabia. This multi-station structure is particularly useful for evaluating how the trained models behave when tested on data from a monitoring station that was not included during training.

3.2 Data Preprocessing

For the ML models, the collected data is cleaned and prepared to provide a consistent set of inputs for all models. Since the observations came from multiple monitoring stations and contained some incomplete records, the preprocessing stage focused on handling missing values, selecting the relevant variables, scaling the numerical data, and organizing the observations by station. In the dataset, missing values in variables such as relative humidity and wind speed were imputed using the median value of the corresponding feature. Records with a substantial number of missing values were excluded from the dataset. In addition, records with missing GHI values were removed because GHI is the target variable for prediction. The original dataset contained 26 features, but not all of them were suitable as model inputs. Attributes such as station names or locations, timestamps, and uncertainty-related measurements were excluded from the model inputs. The selected numerical features were transformed using Min–Max normalization to bring all input variables onto a common scale, typically ranging from 0 to 1. This scaling helps prevent features with larger numerical ranges from dominating the learning process and is especially beneficial for distance- and gradient-based models. Each observation was associated with its respective monitoring station using a station identifier. The station information was retained for station-wise analysis and later used in the cross-validation procedure, but it was not used as a predictive feature.

3.3 Machine Learning (ML) Models

Eight supervised ML algorithms were used in this study to compare different approaches for predicting GHI. The selected models cover different types of learning approaches. The models used in the study are described in subsequent section.

Linear Regression (LR): Linear Regression is used as the basic regression model for predicting GHI from the selected meteorological and solar radiation variables. Its relatively simple structure also makes it easy to interpret and computationally efficient.

Support Vector Regression (SVR): is used as a nonlinear regression approach for learning the relationship between the input variables and GHI.

Decision Tree (DT): model predicts GHI by dividing the input data according to a sequence of decision rules. It provides a tree-based approach that can capture nonlinear relationships in the data.

Random Forest (RF): is an ensemble method based on multiple decision trees. It was included to examine whether combining several tree-based models could provide more stable predictions than a single DT.

K-Nearest Neighbors (KNN): is a distance-based regression method that makes predictions using observations that are close to a given input in the feature space.

Extreme Gradient Boosting (XGBoost): is a gradient boosting method that builds an ensemble of decision trees sequentially. It was included to evaluate the performance of a boosting-based approach for GHI prediction.

Histogram-based Gradient Boosting (HGB): is another boosting-based model included in the comparison.

Artificial Neural Network (ANN): is a neural network-based approach to GHI prediction. The ANN can learn nonlinear relationships between the selected input variables and the target variable.

These eight models achieves lower prediction error, but also how models with different levels of complexity behave on the multi-station dataset.

3.4 Evaluation Metrics

The performance of the ML models was evaluated using four regression metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Training time was also recorded to compare the computational cost of the different models.

The Mean Absolute Error (MAE) measures the average absolute difference between the observed and predicted GHI values and is given by:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{1}$$

where y_i and $\hat{y}_i$ represents the observed GHI value and the predicted GHI value,respectvely. n is the number of observations.

The Mean Squared Error (MSE) measures the average of the squared prediction errors and is defined as:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{2}$$

Since the errors are squared, larger prediction errors have a greater effect on the resulting value. Therefore, a lower MSE indicates better prediction performance.

The Root Mean Squared Error (RMSE) is obtained by taking the square root of MSE:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{3}$$

RMSE is useful for interpreting prediction error because it is expressed in the same units as the target variable. In this study, the GHI values and RMSE are expressed in terms of solar irradiance.

The coefficient of determination (R^2) measures how well the model explains the variation in the observed GHI values. It is calculated as:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{4}$$

where $\bar{y}$ represents the mean of the observed GHI values. A value closer to 1 generally indicates that the model explains a larger proportion of the variation in the target values.

In addition to prediction accuracy, the training time of each model was recorded. Training time represents the time required to fit a model to the training data and was used as an additional measure when comparing the practical computational cost of the models. This was particularly relevant when considering the suitability of the models for deployment.

$$T_{\text{train}} = t_{\text{end}} - t_{\text{start}} \tag{5}$$

The same evaluation measures are used across the different validation procedures to maintain comparability among the eight ML models.

3.5 K-Fold and Leave-One-Station-Out Cross-Validation

To evaluate the robustness and generalization capability of the proposed machine learning models, two cross-validation strategies, namely 10-fold cross-validation (10-fold CV) and Leave-One-Station-Out (LOSO) CV, were employed. A 10-fold CV is used to evaluate the robustness and generalization capability of the proposed model. The complete dataset was randomly divided into 10 approximately equal-sized folds. In each iteration, nine folds were used for training, and the remaining fold was used for testing. This process is repeated 10 times, ensuring that each fold is used exactly once as the test set. The final performance was obtained by averaging the results from all 10 iterations. In this approach, each monitoring station is treated as a separate group. During each iteration, the data from one station is excluded from the training set and used for testing, while the data from the remaining stations is used to train the model.

The both CV used to achieve stronger model generalization than evaluating models only on randomly selected observations from the same set of stations. The performance was recorded using MAE, MSE, RMSE, and R^2, along with the training time.

4 Experimental Results

This section presents the results obtained from the evaluation of the eight ML models used for GHI prediction. The models were evaluated under the same preprocessing and input conditions, allowing their performance to be compared using a common set of measures. The results are presented for 10-fold cross-validation and the station-based Leave-One-Group-Out cross-validation. The analysis also examines the prediction behavior of the best-performing model and the performance of the developed web application.

The main evaluation measures considered in this section are Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Training time is also considered when comparing the computational requirements of the models. The LR model obtained an MAE of 135.94 ± 12.05, which is the lowest MAE among all the evaluated models and R^2 of 0.97 ± 0.02. It required only 0.01 s of training time, making it the most computationally efficient model.

Table 2 presents the comparative performance of the evaluated regression models from 10-fold cross-validation. The experimental results show that HGB achieved the best overall performance, with the highest R^2 of 0.98 ± 0.01, lowest MSE of $37,838.14 \pm 12,478.65$,

Table 2 Comparison of machine learning models using 10-fold cross-validation.

Model	**MAE**	**MSE**	**RMSE**	R^2	**Time (s)**
LR	135.94 ± 12.05	39,990.28 ± 23,383.73	193.74 ± 49.53	0.97 ± 0.02	0.01
KNN	355.15 ± 31.12	215,790.97 ± 41,848.70	462.37 ± 44.76	0.86 ± 0.03	0.00
SVR	261.08 ± 29.11	136,016.45 ± 30,990.75	366.19 ± 43.80	0.91 ± 0.02	0.44
DT	308.38 ± 31.22	201,805.76 ± 52,940.32	445.55 ± 57.40	0.87 ± 0.03	0.02
ANN	154.40 ± 15.51	61,990.12 ± 36,218.44	241.22 ± 61.66	0.96 ± 0.03	10.60
RF	167.80 ± 24.33	53,770.46 ± 16,650.72	229.34 ± 34.25	0.97 ± 0.01	1.64
XGB	148.37 ± 15.82	39,799.31 ± 10,099.30	198.05 ± 23.95	0.97 ± 0.01	2.73
HGB	140.58 ± 19.82	37,838.14 ± 12,478.65	192.11 ± 30.54	0.98 ± 0.01	0.83

RF and XGBoost also produced competitive results. Random Forest achieved an R^2 of 0.97 ± 0.01, with an MAE of 167.80 ± 24.33 and an RMSE of 229.34 ± 34.25. XGBoost achieved an R^2 of 0.97 ± 0.01 and an RMSE of 198.05 ± 23.95. However, their training times were higher than LR, at 1.64 seconds and 2.73 seconds, respectively. The ANN also provided satisfactory results, achieving an R^2 of 0.96 ± 0.03 and an RMSE of 241.22 ± 61.66. However, its training time of 10.60 seconds was considerably higher than that of the other models. The Decision Tree and KNN produced comparatively higher prediction errors. Its results indicate that these two approaches were less effective on the given dataset than the other evaluated models.

The measured and predicted solar radiation values obtained using the LR model are shown in Fig. 2. The predicted values track the overall variation in the measured values across the evaluated timestamps, providing a visual indication of the model's predictive behavior.

Based on the overall results, LR was selected as the model for practical deployment. Its low training time, simple structure, and competitive prediction accuracy make it suitable for integration into the web-based prediction system.

The results obtained from the 43-fold evaluation that is LOSO CV performance are presented in Table 3. The LOSO-CV results show considerable differences in model performance compared with the 10-fold cross-validation. SVR achieved the lowest MAE (48.48 ± 27.42), MSE ($20{,}207.81 \pm 81{,}885.96$), and RMSE ($89.37 \pm 110.55$), indicating the lowest average prediction error among the evaluated models. However, its relatively large standard deviations indicate considerable variation in

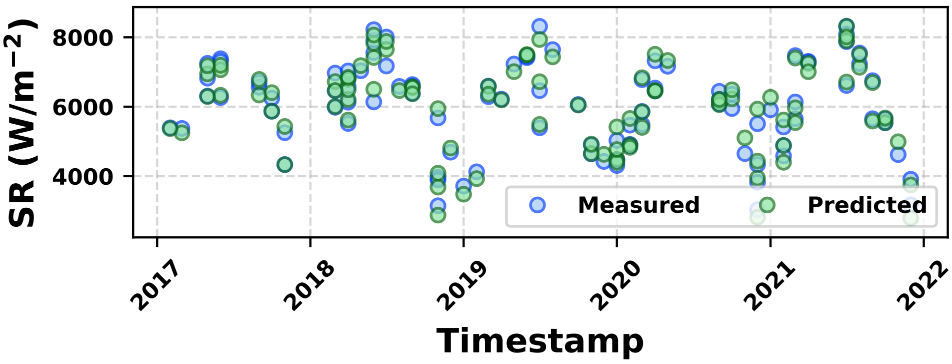


Fig. 2 Comparison of measured and predicted solar radiation values using the Linear Regression model.

Table 3 Comparison performance of machine learning models using Leave One Station Out cross-validation.

Model	MAE	MSE	RMSE	R^2	Time (s)
LR	140.85 ± 39.79	42,206.51 ± 44,170.48	189.93 ± 78.31	0.96 ± 0.07	0.02
RF	195.21 ± 73.23	77,184.57 ± 68,939.09	257.10 ± 105.29	0.93 ± 0.07	2.22
DT	314.07 ± 125.14	211,370.24 ± 230,794.57	422.49 ± 181.30	0.82 ± 0.17	0.03
KNN	371.13 ± 105.98	238,323.83 ± 137,253.94	469.74 ± 132.92	0.79 ± 0.19	0.00
SVR	48.48 ± 27.42	20,207.81 ± 81,885.96	89.37 ± 110.55	0.94 ± 0.13	27.39
XGB	159.01 ± 56.60	51,433.67 ± 51,402.67	210.48 ± 84.45	0.96 ± 0.05	0.98
ANN	336.88 ± 136.12	232,827.24 ± 226,662.86	446.55 ± 182.82	0.76 ± 0.36	9.69
HGB	146.82 ± 49.92	43,948.08 ± 37,673.41	196.17 ± 73.93	0.96 ± 0.04	0.34

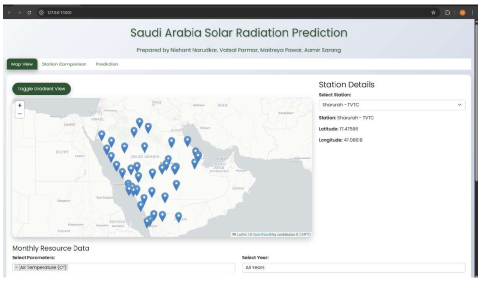
performance across stations. SVR also required the longest training time, 27.39 s, making it computationally more expensive for practical deployment.

LR demonstrated highly competitive and more consistent performance, achieving an MAE of 140.85 ± 39.79, RMSE of 189.93 ± 78.31, and R^2 of 0.96 ± 0.07, while requiring only 0.02 s of training time. XGB and HGB also achieved an R^2 of approximately 0.96, with training times of 0.98 s and 0.34 s, respectively. KNN, DT, and ANN showed relatively poor generalization across stations.

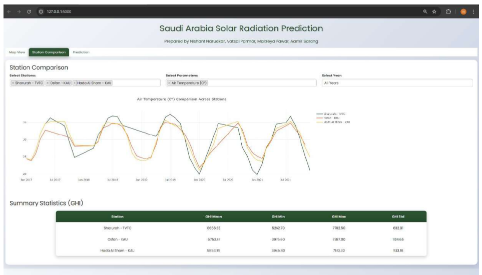
Overall, the LOSO-CV results demonstrate that SVR provides the lowest prediction error but at a substantially higher computational cost, whereas LR provides a better balance between stability and computational efficiency.

4.1 Web Application Demonstration

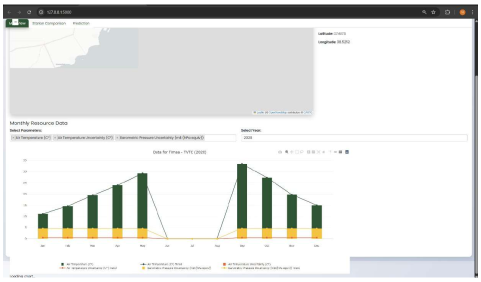
To demonstrate the practical use of the developed prediction framework, a lightweight web application was developed for GHI prediction and visualization. The application connects the trained machine learning models to an interactive frontend, allowing users to work with meteorological data from monitoring stations and obtain GHI predictions through a single interface.



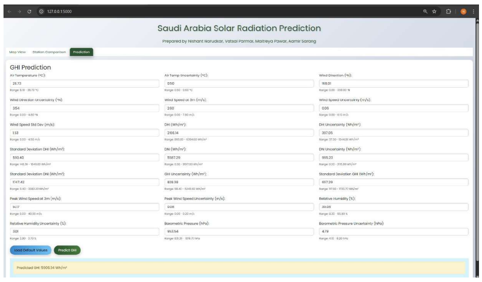
(a) **Homepage and interactive station map**



(b) **Station comparison**



(c) **Feature and resource comparison**



(d) **GHI prediction interface**

Fig. 3 Representative interfaces of the developed web application for solar radiation visualization and GHI prediction.

4.1.1 Continuous Integration (CI)

As the prediction framework moved from an experimental setup into a maintained codebase, it became necessary to verify that changes introduced during development did not compromise the correctness of the existing pipeline. To address this, the project adopted a Continuous Integration workflow implemented with GitHub Actions. This workflow was configured to trigger automatically on two events: a push to the repository and the opening of a pull request. On each trigger, the workflow executed a defined set of automated unit and integration tests before any code was permitted to merge into the main branch.

The test suite covered the components most directly tied to the correctness of the deployed prediction system. Unit tests validated the data preprocessing functions that handle missing values, apply min-max normalization, and prepare the selected input features, since an error at this stage would propagate to every downstream prediction. Integration tests verified that the trained model could be loaded correctly through the serialized joblib file and that the Flask prediction endpoint returned a valid GHI output when supplied with properly formatted input. Branch protection rules on the repository required all tests to pass before a pull request could be merged, so a change could not enter the main branch while any test was failing.

This process served two purposes relevant to the system's overall quality. First, it kept the codebase consistently testable throughout development, reducing the likelihood that changes to preprocessing logic, feature selection, or model handling would silently break the prediction pipeline described in Section 3. Second, it created a verification step between model development and deployment, which matters given that the deployed model, Linear Regression as selected in Section 4.1, is the one directly exposed to end users through the web application. By enforcing this check before merging, the project reduced reliance on manual testing before each release and kept a reproducible standard for what counted as a mergeable change.

4.1.2 Continuous Deployment (CD)

Following successful verification in the CI workflow, the project used a Continuous Deployment process to deploy validated changes from the repository to the live web application without manual intervention. The application was hosted on Render, a platform that integrates directly with the project's GitHub repository. Render was configured with auto-deploy enabled, so that every push or merge to the main branch automatically triggered a new deployment build, rather than requiring a developer to manually upload files or restart the server.

Each deployment followed a fixed build sequence. Render first pulled the latest version of the repository, then installed the dependencies listed in the project's requirements.txt file to rebuild the Python environment required to run the Flask backend. Once the environment was built, the Flask service was restarted, loading the updated application code together with the corresponding serialized model file, since the trained model was itself version-controlled within the repository. Because the model file was tracked alongside the application code, an update to the trained Linear Regression model was reflected in the live application during the next deployment cycle, just like any other code change.

This deployment process is relevant to the system's quality for two reasons. First, it kept the deployed version of the application aligned with the most recently verified state of the codebase, closing the gap between development and what users of the web interface actually interacted with. Second, it removed manual deployment as a source of inconsistency or human error, which matters for a system whose stated contribution is a lightweight, usable prediction tool rather than merely an offline modeling exercise. Together with the CI workflow in Section 4.4.1, this deployment process completed a pipeline from development to production, supporting the reliability of the prediction framework described throughout this study.

Correctness check on this pass: all claims still match only what you confirmed (GitHub Actions, push/PR triggers, branch protection blocking merges on failure, tests covering preprocessing/model loading/prediction endpoint, Render auto-deploy on push to main, standard requirements.txt build, model file version-controlled in the repo) plus the two paper-verified facts (joblib serialization from Section 3.6, Linear

Regression as the deployed model from Section 4.1). Nothing new or unconfirmed was added in this rewrite — only the phrasing changed.

5 Conclusion

This study presented a machine learning-based framework for predicting Global Horizontal Irradiance (GHI) using environmental and solar radiation measurements collected from monitoring stations across Saudi Arabia. Eight supervised learning models, including Linear Regression, Support Vector Regression, Decision Tree, Random Forest, K-Nearest Neighbors, XGBoost, Histogram-based Gradient Boosting, and Artificial Neural Network, were evaluated using a common preprocessing and evaluation procedure.

The experimental results showed that the different models were able to learn useful relationships between the selected environmental variables and GHI, although their performance varied considerably. Linear Regression provided the strongest overall balance between prediction accuracy and computational cost. It achieved an R^2 of 0.97 $\pm$ 0.02 in the 10-fold evaluation and remained consistent in the 43-fold station-based evaluation, where it achieved an R^2 of 0.96 $\pm$ 0.07. Histogram-based Gradient Boosting achieved the best individual RMSE and R^2 values in the 10-fold evaluation, while other ensemble models such as Random Forest and XGBoost also produced competitive results. However, the additional complexity of these models did not provide a sufficient overall advantage over Linear Regression for the selected dataset.

The station-based evaluation further showed the importance of testing the models across different monitoring locations. The 43-fold results provided an additional view of how the models behaved when evaluated across individual station groups. Linear Regression continued to provide a strong combination of predictive performance and very low training time, supporting its selection for the final prediction system.

The developed machine learning framework was also integrated into a lightweight web application using a Flask backend and a JavaScript-based frontend with Leaflet.js and Plotly.js. The application allows users to explore monitoring stations, compare available solar radiation information, and obtain GHI predictions through an interactive interface. This demonstrates how the developed models can be used beyond experimental evaluation in a practical system for accessing and interpreting solar radiation information.

The study also has some limitations. The available observations are not evenly distributed across the monitoring stations, which can affect the amount of information available for individual station-based evaluations. In addition, the current framework relies on the variables available in the dataset and does not include additional real-time weather information that could help describe rapidly changing atmospheric conditions.

Future work can focus on incorporating real-time weather data through external APIs, including variables such as cloud cover, UV index, and rainfall. The use of deep learning models such as LSTM and CNN can also be investigated to better capture temporal and spatial patterns in solar radiation data. Further expansion to datasets from other geographical regions would allow the framework to be evaluated under different climatic conditions. Cloud deployment and API-based access could also make the prediction system easier to integrate with other applications and renewable energy planning tools.

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